BankPulse AI: D2 Submission

Data Audit Report, Datasheet and Versioned Dataset Package

Early-Warning Analytics for Bank Financial Health

Item	Detail
Course / milestone	ANCP Term 5, 2026. D2: Data Audit, Datasheet, Versioned Dataset
Team	Team 1 (T01)
Members	Chahatleen Kaur, George Thomas, Jigyas Niraj Borah
D2 lead	George Thomas (data audit); compiled with Jigyas Niraj Borah
Dataset version	bank_financial_data_v3.csv (v3)
Audit date	22 October 2026
Coverage	20 banks; reporting periods 2017-03-31 to 2026-09-30

1. Executive Summary

BankPulse AI ranks the most unusual bank-year observations using an Isolation Forest, compared with a statistical baseline, and produces an analyst-reviewed advisory report (D1 charter). It is anomaly detection to support review. It does not predict failure or rate any bank. D2 audits the data that will feed that work.

The team consolidated 20 bank Excel workbooks into one long-format table, bank_financial_data_v3.csv, with 7,061 records and six columns across four statement sections: Profit & Loss, Quarterly Results, Balance Sheet and Cash Flow. The audit finds the structure sound: no non-numeric values, no duplicate keys, no conflicting records. Missing values are kept as missing and never read as zero. The data comes from Screener.in and Bloomberg. Provenance, licence and outlier review are the main open risks, and they are stated openly in Sections 2.2, 2.3 and 6.



Figure 1. D2 work process, from raw workbooks to D2 outputs.

Raw files are the original bank workbooks. Extraction reads the financial figures into one format. Cleaning standardises the data and removes exact repeats. Validation checks that the resulting table follows the expected structure, and the D2 outputs are the files kept as evidence.

Audit item	Result
Records in v3	7,061
Banks	20
Distinct reporting periods	18 (2017-03-31 to 2026-09-30)
Columns	Bank, Section, Reporting_Period, Metric, Value, Source_File

Audit item	Result
Exact duplicate rows removed	200 (one copy of each retained)
Conflicting records flagged	0
Quality issues retained in log	1,539
Non-numeric values in Value	0
Duplicate keys after cleaning	0

1.1 Alignment with the D1 charter

D1 said its success targets would be confirmed after this audit. The table shows where the dataset stands against each.

D1 commitment	D2 finding
At least 95% data completeness, if feasible	Missing values are logged (1,539 issues) and retained, never imputed. The 95% target will be applied to the core fields chosen for analysis, and any exclusions will be documented.
100% traceability of ratios and flags	Every record carries Source_File. All values are in INR crore (Section 2.5), so each derived ratio can be traced to its inputs.
50 to 200 bank-year observations	20 banks over multiple annual periods between 2017 and 2026, consistent with D1's expected range.
Ratio families: profitability, asset quality, capital and liquidity, efficiency, growth	v3 holds raw statement line items with no derived ratios. Section 2.6 lists which families the data can support and which need an added source.
Licensing confirmed first	Both sources are licensed or terms-bound. Use is limited to this course and data is not redistributed (Section 2.2).
Primary source Bloomberg	D1 named Bloomberg as the primary source. The team also used Screener.in financial statement exports to reach 20 banks. This is a documented change from D1.

2. Part A: Data Audit

2.1 Scope and structure

Each row is one bank, statement section, reporting period, metric, numeric value and source workbook. The long format supports filtering and aggregation by any of these.

Field	Meaning	Validation
Bank	Standardised bank name	Required; no missing values in final validation
Section	Statement area	One of PROFIT & LOSS, QUARTERS, BALANCE SHEET, CASH FLOW
Reporting_Period	Period-end date	Text, YYYY-MM-DD; annual and quarterly dates both occur
Metric	Financial line item	Labels retained from the source workbook
Value	Numeric value	Zero non-numeric values in final validation
Source_File	Originating workbook	Preserves traceability

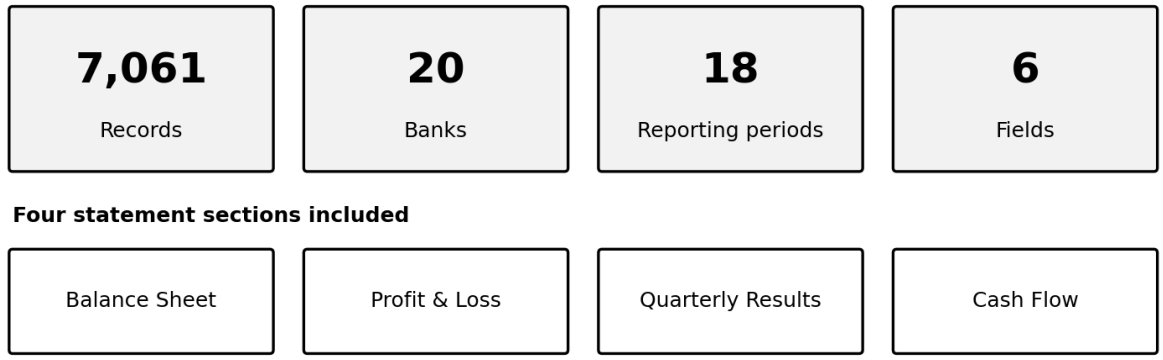


Figure 2. Dataset at a glance and the four statement sections included.

The 7,061 is the number of data rows, not banks. The dataset holds information from 20 banks across 18 reporting dates, arranged in six columns. The four sections are Balance Sheet (what a bank owns and owes), Profit & Loss (income and expenses), Quarterly Results (recent quarter performance) and Cash Flow (movement of cash). The figure shows which sections are included, not how many records each holds.

2.2 Source and access

Raw inputs were 20 Excel workbooks, one per bank, held in a project folder named Bank_Data. Sheets included Profit & Loss, Quarters, Balance Sheet, Cash Flow, Customization and Data Sheet. Data was drawn from Screener.in (company financial statement exports) and Bloomberg.

Banks in the dataset (19 shown here): AU Small Finance Bank, Axis Bank, Bandhan Bank, Bank of Baroda, Bank of India, Bank of Maharashtra, Canara Bank, Federal Bank, HDFC Bank, ICICI Bank, IDBI Bank, IDFC First Bank, Indian Bank, IndusInd Bank, Kotak Mahindra Bank, Punjab National Bank, SBI, Union Bank, Yes Bank. The complete list of 20 is in the Bank column of the CSV.

Provenance item	Status
Publishers	Screener.in and Bloomberg
Method	Excel workbooks, one per bank, from Screener.in exports and Bloomberg
Date accessed	Files were processed on 22 Oct 2026 (audit date). Original download dates are not recorded.
Licence	Neither source is open data. Screener.in terms of use and the Bloomberg licence apply. Data is used only for this course project and the CSV is not redistributed. Submission to faculty is for assessment only.
Checksum	SHA-256 in Section 4.3

2.3 Quality audit and treatment

Issue	Evidence	Treatment and rationale
Exact duplicates	200 duplicate rows found and removed (cleaning log)	Keep one copy; avoids double counting without discarding unique records
Missing values	1,539 quality issues logged; many line items blank	Kept as missing, never zero. A blank may mean not reported, not applicable or unavailable
Conflicting records	0 in v3; review sheet retained	Preserved as evidence
Non-numeric values	0 in final validation	Only values that parse reliably are converted;

Issue	Evidence	Treatment and rationale
		failures are logged
Structurally unavailable metrics	Raw Material Cost, Change in Inventory and Power and Fuel are blank in inspected sheets	Treated as not applicable to banks and excluded from modelling; no imputation
Outliers	Outlier review not completed at D2; stated as a limitation	Extreme values are checked against source statements and never deleted automatically

2.4 Representativeness and bias

Potential bias	Why it matters	Control
Bank coverage	20 banks are not the whole Indian banking sector	Described as a selected sample, not a census
Selection	The 20 banks were taken as the top 20 banks of India, as of the audit date	Rationale stated; current-ranking selection introduces survivorship bias, noted
Period mix	Annual and quarterly records coexist; recent quarters are sparser	Modelling uses the annual panel only; quarterly data is used for context
Bank type and size	Large public and private banks differ from small finance banks	Peer-median comparisons by group, not raw cross-bank values
Metric coverage	Some line items are irrelevant or inconsistently reported	Ratios computed only where inputs exist

2.5 Units and scale

Cross-bank comparison needs a common unit. All values in all workbooks are in INR crore, so no scale conversion is needed. Ratios and growth rates are unit-free.

2.6 Feature engineering and leakage risk

At D2 the dataset holds raw line items and no derived features. The planned unit of analysis is one bank-year observation built from the annual sections (Profit & Loss, Balance Sheet, Cash Flow). The data has no outcome labels. Features planned for later stages, and what the current data can support, are below.

D1 ratio family	Planned features	Data support in v3
Profitability	ROA, ROE, net profit margin	Supported if net profit, total assets and equity are present
Asset quality	Gross NPA, net NPA, provisions	Not confirmed at D2; treated as a limitation if the workbooks do not contain these items
Capital and liquidity	Capital adequacy, Tier 1, liquidity	Not confirmed at D2; same treatment
Efficiency	Cost-to-income	Supported if operating expense and income lines are present
Growth	Year-on-year change in assets, deposits, advances	Supported from annual periods; guard against zero denominators
Peer difference	Bank value minus peer median	Computed per year within bank-type groups

Leakage control: each feature uses only the observation year and prior years, and no outcome or future information is used as an input.

2.7 Consent, purpose limitation and responsible use

The fields are corporate financial metrics and bank names, not personal data, so no consent is needed. Under the DPDP Act, 2023, the dataset is not designed to hold personal data. The team will add none, will limit use to the stated academic purpose, will restrict access to working files, and will follow the Screener.in and Bloomberg terms. This is a project-level statement, not a legal opinion. Outputs flag unusual observations for analyst review. They are not investment advice and do not declare any bank distressed.

3. Part B: Datasheet (bank_financial_data_v3.csv)

3.1 Motivation

The dataset supports analyst review of unusual financial patterns across 20 banks over multiple years, and the peer benchmarking, baseline and Isolation Forest work planned in D1. It is not a live feed and does not cover all Indian banks.

3.2 Composition

Attribute	Description
Version	v3, bank_financial_data_v3.csv
Records / fields	7,061 records; 6 fields
Banks / periods	20 banks; 18 periods, 2017-03-31 to 2026-09-30
Sections	PROFIT & LOSS, QUARTERS, BALANCE SHEET, CASH FLOW
Format	CSV, long format, one record per bank, section, period, metric
Units	INR crore (all workbooks)
Traceability	Source_File names the originating workbook

3.3 Collection and preprocessing

Bank workbooks were read and converted into the standard six-column format. Source filenames were kept, quality issues logged, and exact duplicates removed with one copy retained. Raw workbooks are preserved unchanged.

Check	Result
Standard structure	Six fixed columns
Duplicates	200 exact duplicates removed
Missing values	1,539 issues logged; never filled with zero
Conflicts	0 reported; review sheet retained
Numeric validation	0 non-numeric values
Outliers	Review not completed at D2

3.4 Uses, non-uses and limitations

Intended uses: anomaly detection to support analyst review, peer benchmarking, trend analysis, and carefully defined ratios.

Not intended for: investment decisions, failure or distress prediction, representing all Indian banks, treating missing values as zero, or mixing annual and quarterly values without normalisation.

3.5 Distribution and maintenance

The cleaned dataset is stored in the project repository with a version tag, data dictionary, cleaning script, cleaning log and checksum. Earlier versions are never overwritten. The data is refreshed only when a new source set is collected and validated. Editing rights are limited to team members. Because the data is terms-bound, it is not shared outside the course.

4. Part C: Versioned Dataset Package

4.1 File manifest

File	Role
bank_financial_data_v3.csv	Final cleaned dataset (7,061 rows, 6 columns); frozen submission copy
D2_Final_Dataset_Working_v3.xlsx	Working and audit workbook with data sheets, Data_Quality_Log and Conflict_Review
D2_Data_Dictionary.xlsx	Definitions of the 6 fields
D2_Cleaning_Log_v3.csv	Duplicate removals and actions
Cleaning script	Recreates v3 from the raw workbooks

4.2 Version log

Version	Change summary
v1, v2	Earlier working versions, superseded by v3
v3	Exact duplicates removed (200); quality log and conflict review added; final validation passed. 7,061 records

Rule: the version number increases whenever content or cleaning logic changes, and earlier versions are never overwritten.

4.3 Checksums

A SHA-256 checksum identifies the exact frozen file. It is generated for the frozen CSV and the cleaning script using:

Linux / macOS: sha256sum bank_financial_data_v3.csv

Windows PowerShell: Get-FileHash bank_financial_data_v3.csv -Algorithm SHA256

4.4 Reproducibility

Reproducibility test: a second team member runs the cleaning script on the raw workbooks into a clean folder and compares row count (7,061), column order, sorted contents and checksum with the frozen file. This independent run is not claimed in this submission, so exact reproducibility is stated as a limitation.

5. Validation Summary

Check	Result
Record count	7,061
Bank count	20
Reporting periods	18

Check	Result
Field count	6
Duplicate keys after cleaning	0
Non-numeric Value entries	0
Conflicts	0

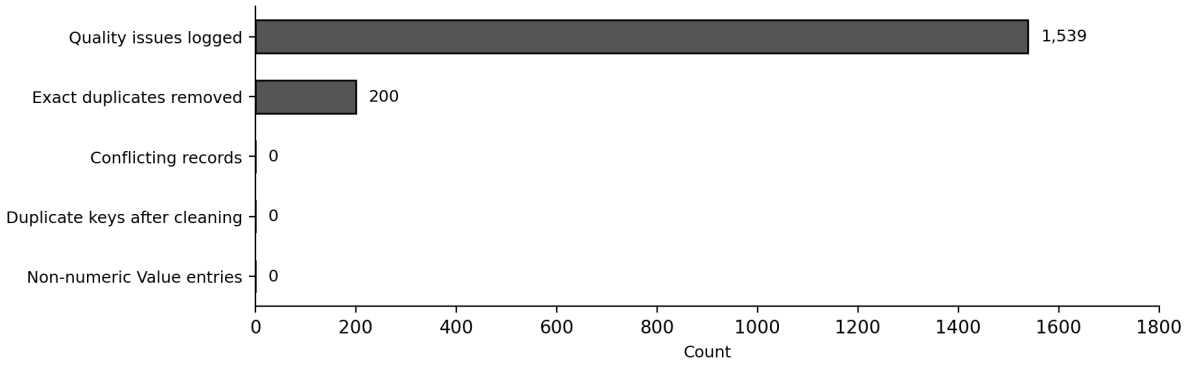


Figure 3. Cleaning and validation counts for v3.

These checks confirm structure. They do not prove that every metric exists for every bank and period.

6. Limitations

The main limitations are as follows. Data from Screener.in and Bloomberg is terms-bound: sharing is restricted and use is limited to this course. The dataset is a selected sample of 20 banks, not the whole sector, and there has been no completed outlier review and no independent reproduction run at D2. Annual and quarterly data are mixed, so analysis uses the annual panel only. Metric definitions may differ between the two sources and between banks, although units are uniform (INR crore). Several metrics are structurally blank for banks and are excluded. The data is raw statement items only, with no derived ratios at this stage, and it has no outcome labels, so results are anomaly flags for review, not predictions.

7. Conclusion and Next Step

The team has a structured, versioned, traceable bank-financial dataset with its cleaning and quality evidence preserved. The open risks are stated in Section 6.